VIEW-IMPOSTOR AUDITS FOR SPARSE-VIEW 3D SCENE WORLD MODELS

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ABSTRACT

Learned 3D scene world models increasingly represent futures as neural fields, Gaussian splats, point clouds, or occupancy volumes. Their deployment interface often samples several candidate futures and keeps the one with the strongest sparse-view proxy score. This paper audits a specific geometry failure in that interface. Sparse projective scores can be blind to hidden occupancy, so larger candidate pools make it easier to find a view impostor: a scene that matches the scored views while disagreeing with unobserved 3D structure. We build a deterministic voxel benchmark that exposes this mechanism. The naive sparse-view selector raises proxy score from 0.791 at $N = 1$ to 1.000 at $N = 64$, while true 3D IoU falls from 0.520 to 0.340 and hidden-region error rises from 0.442 to 0.929. A reranker using ensemble geometry consensus, visual-hull volume plausibility, and surface-complexity penalties recovers much of the hidden geometry, reaching 0.888 true IoU at $N = 16$. The contribution is a scoped audit protocol for candidate scene-world models, not a universal claim about all neural renderers.

1 INTRODUCTION

Candidate scene inference is attractive for world models: sample several plausible future scenes, score them with a task proxy, and execute or report the best. In 3D scene representations this proxy often comes from sparse observations: rendered views, depth maps, partial point clouds, or task-relevant camera rays. The proxy is cheaper and more available than full 3D ground truth, but it is not injective. Many different 3D scenes can share the same silhouettes and visible front surfaces.

This paper asks what happens when that ambiguity becomes a selection interface. The answer in our controlled benchmark is uncomfortable: increasing the number of sampled candidates can make the selected scene look better under the sparse proxy while making it worse as a 3D scene. The selector finds rare view impostors. These candidates are excellent under the scored views and poor in hidden regions.

Our contribution is a mechanism-specific diagnostic rather than a new renderer:

- a synthetic 3D scene-world benchmark with primitive occupancy scenes, stochastic candidate futures, sparse orthographic proxy views, and hidden geometry labels for evaluation;
- diagnostics for proxy score, true 3D IoU, hidden-region error, selected-candidate diversity, view-impostor rate, and the geometric exploitation gap;
- a construction-level proposition showing why sparse-view equivalence enables candidate-pool amplification;
- a simple coverage-aware reranker that reduces the exploitation gap in the benchmark without accessing hidden ground truth.

2 RELATED WORK

Neural scene representations. NeRF Mildenhall et al. (2020), Scene Representation Networks Sitzmann et al. (2019), sparse-view NeRF methods such as pixelNeRF Yu et al. (2021), IBRNet Wang et al. (2021), RegNeRF Niemeyer et al. (2022), and SparseNeRF Wang et al. (2023) study how to represent or infer scenes from limited views. Gaussian Splatting Kerbl et al. (2023) and follow-up geometry-aware splatting methods such as SuGaR Guedon and Lepetit (2024) and 2DGS Huang et al. (2024) provide explicit, fast renderable scene representations. These papers motivate the architecture class, but they do not isolate sparse-view candidate selection over scene futures.

Occupancy and scene world models. Occupancy Networks Mescheder et al. (2019), Convolutional Occupancy Networks Peng et al. (2020), MonoScene Cao and de Charette (2022), VoxFormer Li et al. (2023), and OccWorld Zheng et al. (2024) address 3D completion and occupancy prediction. Our benchmark uses voxels because they make hidden-region error directly measurable; the question is not how to build the best occupancy predictor, but how a sparse proxy behaves when choosing among multiple predicted occupancies.

Candidate selection and proxy overoptimization. Multi-hypothesis prediction and best-of-many objectives Guzman-Rivera et al. (2012); Bhattacharyya et al. (2018) encourage diverse outputs. Reward and verifier literature shows that selecting against an imperfect proxy can overoptimize that proxy Cobbe et al. (2021); Lightman et al. (2023). Our work specializes this concern to 3D projective ambiguity: sparse views create equivalence classes of hidden geometry.

Classical multiview geometry. Visual hull and space-carving results Laurentini (1994); Kutulakos and Seitz (2000) already show that sparse views underdetermine shape. We use this as the mechanism, not as a novelty claim. The new empirical question is how often sparse-view candidate selection finds such ambiguous impostors as the candidate pool grows.

3 BENCHMARK

Scenes are binary occupancy grids composed of random boxes and ellipsoids. A target scene is treated as the future scene to be predicted. A simulated world-model sampler emits N candidate scenes from three modes:

- **faithful**: small shifts, deletions, and local additions around the target;
- **drift**: larger global shifts and noise;
- **view impostor**: copies first visible surfaces under the sparse proxy views and randomizes hidden voxels behind those surfaces.

The sparse proxy observes two orthographic axes. For each view it scores silhouette IoU and first-hit depth agreement:

$$s_{\text{proxy}}(x, y) = \frac{1}{|V|} \sum_{v \in V} 0.7 \, \text{IoU}(\text{sil}_v(x), \text{sil}_v(y)) + 0.3 \, \text{depthsim}_v(x, y).$$

Evaluation uses true 3D IoU and hidden-region IoU, where the hidden region is the target visual hull excluding voxels visible from the proxy views.

4 MECHANISM

Proposition 1 (View-impostor amplification)**.** *For a finite sparse set of projection views, there exist distinct 3D occupancies with identical proxy score and different true 3D IoU. If a sampler independently produces a proxy-dominating hidden-inconsistent candidate with probability $p > 0$, then the probability that a sparse-view selector sees at least one such candidate among N samples is $1 - (1 - p)^N$.*

Sketch. Copy the first occupied voxel on every scored ray from a target scene. Any occupancy added or removed strictly behind those copied first hits does not change the scored silhouettes or first-hit depths. Therefore sparse proxy score cannot distinguish the target from these hidden variants. Under independent sampling, the probability of at least one hidden-inconsistent variant among N samples is the complement of sampling none. If that variant has higher proxy score than faithful alternatives, the selector chooses it.

The proposition is deliberately narrow. It assumes binary occupancy, orthographic first-hit views, and independent samples. Its purpose is to justify the diagnostic tested below, not to prove a universal theorem for all renderers.

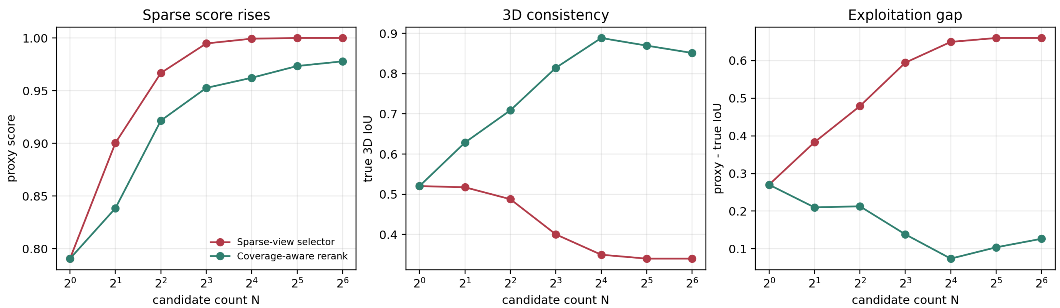

Figure 1: Naive sparse-view selection improves the proxy while reducing true 3D consistency. The coverage-aware reranker trades a small amount of proxy score for much better 3D IoU and lower exploitation gap.

Table 1: Full synthetic run with 72 scenes. Values are means over scenes.

method	N	proxy	true IoU	hidden error	gap
naive	1	0.791	0.520	0.442	0.270
naive	8	0.995	0.400	0.841	0.594
naive	16	0.999	0.350	0.915	0.650
naive	64	1.000	0.340	0.929	0.660
repaired	1	0.791	0.520	0.442	0.270
repaired	8	0.953	0.814	0.200	0.138
repaired	16	0.962	0.888	0.121	0.074
repaired	64	0.978	0.851	0.183	0.127

5 Repair

The repair keeps the candidate set fixed and changes only the selection score:

$$s_{\text{repair}} = 0.58 s_{\text{proxy}} + 0.27 s_{\text{consensus}} + 0.22 s_{\text{volume}} - 0.12 s_{\text{surface}}.$$

The consensus term is IoU with a thresholded ensemble occupancy. The volume term penalizes candidates whose occupied mass is implausible under the visual hull induced by the sparse views. The surface term penalizes thin shell-like occupancy. No term uses hidden ground-truth voxels.

6 Results

The headline result is the divergence between proxy and true 3D consistency. For the naive sparse-view selector, proxy score reaches 1.000 at $N = 32$ and $N = 64$, while true IoU falls to 0.340. Hidden-region error rises to 0.929. The selected candidate is a view impostor in 100% of scenes at $N = 32$ and $N = 64$.

The repair is not perfect, but it targets the mechanism. It reaches 0.888 true IoU at $N = 16$ and keeps the exploitation gap at 0.074. At $N = 64$, the repair still selects some impostors, but true IoU remains 0.851, far above the naive 0.340.

7 Limitations

This is a synthetic mechanism study. The candidate sampler is simulated, the proxy is a simplified silhouette/depth score, and the repair is hand-specified. The benchmark does not establish failure rates for real NeRF, Gaussian splatting, point-cloud, or autonomous-driving occupancy systems. The next submission-quality step is to attach the diagnostic to a real neural-scene generator and evaluate with held-out 3D reconstructions or multi-view captures.

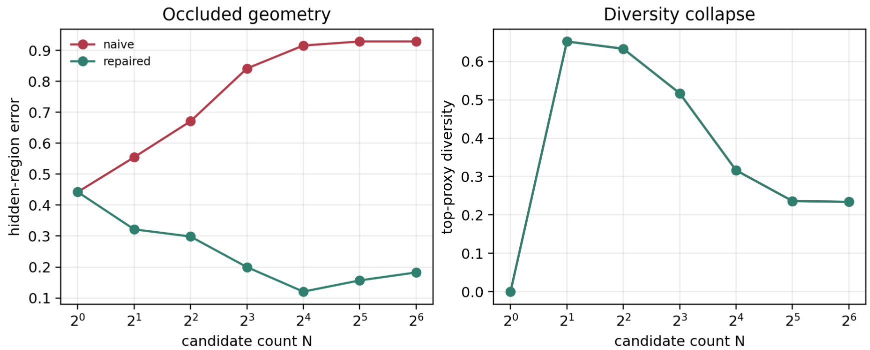

Figure 2: Naive selection increases hidden-region error and collapses top-proxy diversity as N grows.

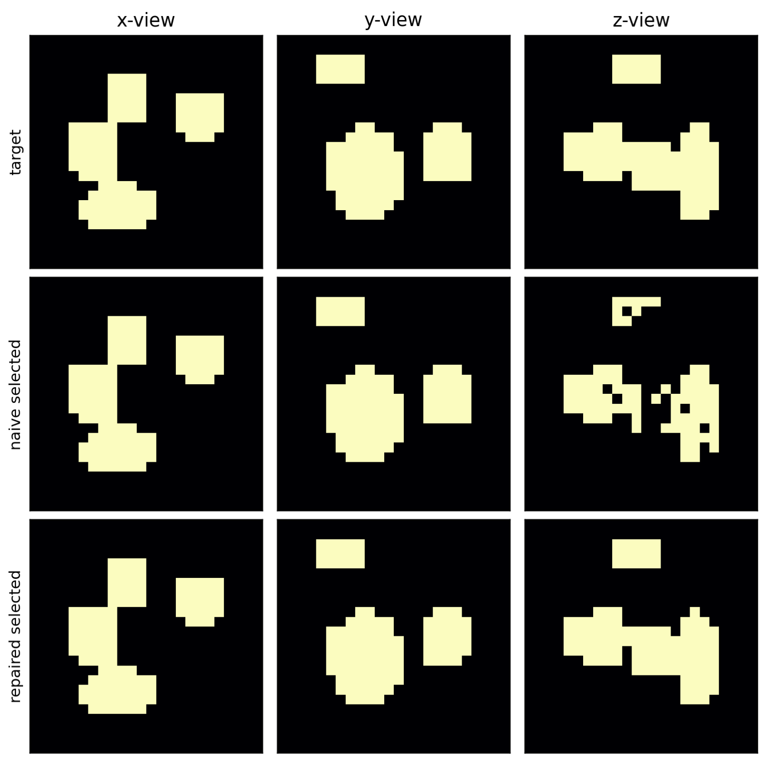

Figure 3: Example projections. The naive selected candidate matches sparse scored views but differs in unscored geometry; the repaired candidate preserves more global occupancy.

8 Reproducibility

The repository contains deterministic seeds, tests, smoke and full presets, CSV tables, and generated figures. The main commands are:

```
python -m pytest
python experiments/run_synthetic.py --preset smoke --output results/smoke
python experiments/run_synthetic.py --preset full --output results/full
python scripts/build_paper.py
```

9 ICLR CHECKLIST

Anonymous submission. The paper and repository use anonymous author metadata.

Claims. The claims are limited to the synthetic benchmark and construction-level mechanism.

Code and data. All benchmark code is included. Data are generated procedurally.

Compute. Experiments run on CPU. The full preset uses 72 scenes and finishes in minutes on a standard laptop.

Limitations. The main limitation is lack of real neural-scene experiments.

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